

Correlation Analysis (for Categorical Data)

□ χ^2 (chi-square) test:

$$\chi^2 = \sum_i^n \frac{\overset{\text{observed}}{\downarrow} (O_i - E_i)^2}{\underset{\text{expected}}{E_i}}$$

- Null hypothesis: The two distributions are independent
- The cells that contribute the most to the χ^2 value are those whose actual count is very different from the expected count
 - The larger the χ^2 value, the more likely the variables are related
- Note: Correlation does not imply causality
 - # of hospitals and # of car-theft in a city are correlated
 - Both are causally linked to the third variable: population